

HW 2.4.2: Using Matrices to Model Conics

General Form for all Conics: $Ax^2 + Bxy + Cy^2 + Dx + Ey + F = 0$, the Bxy term creates a rotated conic.

Write the Conic in General Form by using Matrices. On questions 1-5 the conics are not rotated so the Bxy term is not needed. Calculators are allowed.

- 1.) Find the equation for a circle containing $(-1,1)$, $(2,3)$, and $(4,2)$. Remember that on a circle A and C are equal, in order to solve with a Matrix set A and C equal to 1.
- 2.) Find the equation of the circle that passes through $(-3,1)$, $(5,-1)$, $(2,-4)$, and $(0,-4)$. You do not need to use all 4 points.
- 3.) Find the equation for a parabola containing $(1,3)$, $(5,1)$, and $(8,4)$. This is a vertical parabola so the C value is 0, set the A value equal to 1 to solve.
- 4.) Find the equation for a parabola containing $(1,3)$, $(5,1)$, and $(8,4)$. This is a horizontal parabola so the A value is 0, set the C value equal to 1 to solve.
- 5.) Find the equation for an ellipse containing $(2,4)$, $(6,2)$, $(3,3)$, and $(4,8)$. Set the A value equal to 1 and solve for C , D , E , and F .

(Optional) Write the Conic in General Form by using Matrices. On questions 6-9 the conics are rotated so the Bxy term is needed.

6.) Find the equation of the conic that passes through $(7,5)$, $(4,4)$, $(5,7)$, $(14,10)$, and $(19,14)$. Set the A value equal to 1.

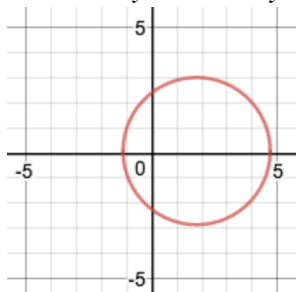
7.) Find the equation of the conic that passes through $(14,22)$, $(10,10)$, $(22,14)$, $(35,25)$, and $(50,50)$. Set the A value equal to 1.

8.) Find the equation of the conic that passes through $\left(\frac{3}{4}, \frac{9}{4}\right)$, $(1,1)$, $\left(\frac{19}{12}, \frac{3}{4}\right)$, $\left(\frac{9}{4}, \frac{3}{4}\right)$, $\left(\frac{33}{5}, \frac{9}{5}\right)$. Set the A value equal to 1.

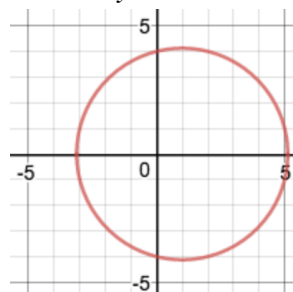
In order to get your answer to match the answer below multiply everything by the Least Common Denominator, which makes the Conic in General Form (all coefficients are integers)

Answers

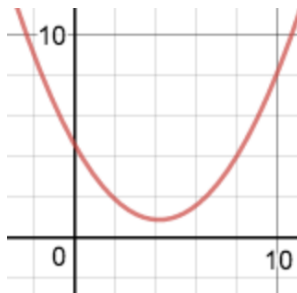
1.) Circle: $7x^2 + 7y^2 - 25x - y - 38 = 0$



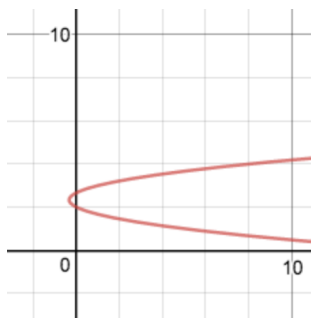
2.) Circle: $x^2 + y^2 - 2x - 16 = 0$



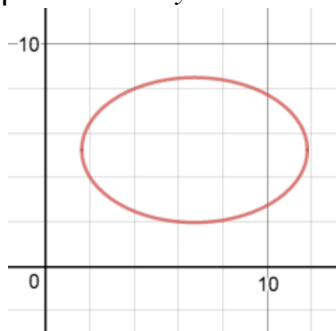
3.) Parabola: $3x^2 - 25x - 14y + 64 = 0$



4.) Parabola: $3y^2 - x - 14y + 16 = 0$



5.) Ellipse: $7x^2 + 17y^2 - 94x - 178y + 600 = 0$

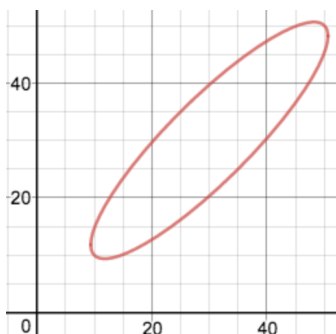


6.) Rotated Parabola: $x^2 - 2xy + y^2 - x - y + 8 = 0$



7.) Rotated Ellipse:

$17x^2 - 30xy + 17y^2 - 120x - 120y + 2000 = 0$



8.) Rotated Hyperbola:

$3x^2 - 10xy + 3y^2 - 4x - 4y + 12 = 0$

